

Sarthak Ajit Kotkar

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SUMMARY

Computer Engineering student (B.E., Pune, graduating 2026) with hands-on experience in **software engineering, full-stack web development, and AI-powered 3D visualization systems**. Skilled in **JavaScript, TypeScript, React, Vue.js, FastAPI, Node.js, and PostgreSQL**, with a track record of building **scalable, data-driven, and interactive applications**. Passionate about creating **intelligent, real-time solutions** that combine **AI, cloud technologies, and modern web frameworks** to solve real-world problems.

SKILLS

Programming Languages: JavaScript, TypeScript, Python, C++, HTML5, CSS3.

Frontend: React.js, Vue.js, Three.js, React Native, Bootstrap, Tailwind CSS.

Backend: Node.js, Express.js, Next.js, FastAPI, REST API, Authentication (JWT, OAuth2).

Database: PostgreSQL, MySQL, MongoDB, Neo4j, Redis, Firestore.

Cloud & DevOps: Google Cloud Platform (Vertex AI, Cloud Run, Firestore, Firebase), Docker, GitHub Actions, Render.

Tools & Platforms: Git, GitHub, Postman, Figma, MongoDB Compass, Replit.

EXPERIENCE

Software Engineer Intern

Equations Work

January 2025 - March 2025, Bavdhan, Pune

- Developed an **AI-powered knowledge graph** using machine learning models to extract and structure insights from complex data.
- Built backend APIs with **Flask** and visualized relationships in 3D using **Babylon.js** and **Neo4j**.

PROJECT

TruthLens - Misinformation Detection WebApp

flash-clover-476914-h6.web.app/

- Developed a **full-stack misinformation detection system** on **Google Cloud Platform** using **React, FastAPI, Cloud Run, and Firebase Hosting**.
- Engineered a **dual-AI backend: Gemini API (DDGS)** for general multi-modal analysis and **Vertex AI (Google Search)** for deep fact-checking.
- Implemented **secure user authentication** with Firebase and stored user data and analysis history in **Cloud Firestore**.
- Deployed a **serverless and scalable architecture** with Cloud Run for APIs and Firebase Hosting for the frontend.
- Tech Stack:** React, FastAPI, Google Cloud Run, Vertex AI, Gemini API, Firebase, Firestore.

3D City Knowledge Graph Visualization

d-city-68274.web.app/

- Designed and implemented a **3D knowledge graph** representing urban infrastructure and interrelationships within a city model.
- Used **Three.js** for real-time 3D visualization and interactive exploration of graph elements.
- Integrated **Firestore(Firebase database)** for data storage.
- Tech Stack:** Three.js, React, Firebase (Firestore, Hosting).

YatriSuraksha - Tourist Safety System

- Engineered a **full-stack safety system** using **FastAPI, React, and WebSockets** for real-time tourist location tracking and visualization on live maps.
- Developed an **AI anomaly detection module** leveraging **MongoDB geospatial queries** to identify risks like geofence breaches, route deviations, and inactivity.
- Built an **incident command center dashboard** with a live map and alert feed to enhance situational awareness for authorities.
- Tech Stack:** FastAPI, React, MongoDB, WebSockets.

EDUCATION

Bachelor of Engineering in Computer Engineering

Marathwada Mitra Mandal College of Engineering • Pune • 2026 • 8.79

CERTIFICATIONS

AWS Academy Graduate - Cloud Foundations

Amazon Web Services Training and Certification • 2025

https://www.credly.com/badges/fa26694c-f430-4776-b54b-7e8967612174/public_url

SQL(Basic) Certificate

Hackerrank • 2024

<https://www.hackerrank.com/certificates/9201c8c4ca1a>